

CSE 6242: Team 102 Project Proposal

1 Introduction

Efficient emergency response is vital to public safety, yet 911 systems often face delays from high call volumes, uneven resource allocation, and complex dispatch dynamics. This project uses **Seattle’s 911 data** to model response performance, optimize unit deployment, and identify operational inefficiencies. By integrating machine learning with interactive visual analytics, we aim to turn raw call data into insights that improve response speed, equity, and system reliability.

2 Literature Review

Recent research emphasizes the growing role of spatiotemporal modeling and visual analytics in improving emergency response and public safety management. Prior studies such as *Space-Time Urban Explorer* [1] and *Optimizing Local Police Practices* [2] demonstrate how interactive dashboards and visual exploration tools can enhance operational insights for law enforcement, yet they lack predictive integration. Building on this foundation, *Spatiotemporal Predictive Geo-Visualization for Real-Time Systems* [3] and *Ada-GCNLSTM* [4] apply deep learning and graph-based methods to forecast spatial crime distributions, achieving high accuracy but at significant computational cost.

Traditional forecasting approaches, including ARIMA, Holt-Winters, and regression models, have also been explored for EMS and 911 call centers [5, 6], showcasing reasonable short-term performance but limited spatial generalization. Recent healthcare analytics research extends these ideas using scalable machine learning pipelines such as Random Forests for near real-time performance prediction [7] and explainable multi-model regression frameworks for emergency forecasting [8], underscoring the importance of model interpretability and continuous retraining. Optimization and clustering studies [9, 10] further highlight data-driven strategies for officer deployment and response time, while online decision-theoretic frameworks [11] and next-generation computer-aided dispatch systems [12] demonstrate how automation can enhance dispatch efficiency. However, administrative and political factors often complicate adoption, as shown in early analyses of CAD implementation [13]. Other works on simulation and algorithm governance [14, 15] emphasize the importance of interpretability and accountability in operational decision support.

These studies highlight the potential of data-driven strategies to improve emergency response and resource allocation. Our project builds on this foundation by integrating lightweight machine-learning models with interactive visualizations to analyze 911 response patterns, bridging predictive analytics and decision-support dashboards for practical public safety insights.

3 Methodology

Our project integrates four analytical models with aligned visualizations to generate actionable insights for Seattle’s 911 operations (Table 1 and Table 2).

With free data, open-source tools, and team-based development, the project incurs no cost. The project targets measurable gains in response speed, equity, and overall system efficiency, aiming for a 10% reduction in high-priority call times through simulation. While risks include data quality and model complexity, the potential payoff is a more efficient, data-driven 911 system that optimizes resources and saves lives.

4 Project Planning

All members will jointly handle pipeline development, code reviews, documentation, and visual design to ensure cross-functional learning and consistency (Table 3). One member, Jiahui Zhang, has dropped out of the course, and the rest of the team (5 members) contributed equally.

Task	Best Model	Outcome
Response Time Prediction	XGBoost / LightGBM	Reduce average dispatch response time through predictive modeling of call priority and location.
Resource Allocation	K-Means Clustering + Reinforcement Learning	Optimize patrol unit deployment and improve service balance across precincts.
Co-Response Effectiveness	XGBoost + Causal Inference	Quantify the operational impact of co-response units on incident clearance times.
Anomaly Detection	Isolation Forest	Identify systemic inefficiencies or abnormal dispatch patterns for process improvement.

Table 1: Summary of analytical models and expected outcomes for the Seattle 911 Triage/Planner project.

Visualization	Description	Insight Provided
Temporal–Spatial Heat Animation	Animated heatmap of call frequency and response time by hour or day.	Reveals city-wide demand cycles and temporal hotspots.
Response Time Decomposition Dashboard	Interactive breakdown of delays by call type, priority, and district.	Identifies key bottlenecks and high-delay categories.
Patrol Coverage Map (Before vs. After)	Side-by-side maps comparing current vs. optimized resource allocation.	Shows the spatial impact of optimization on coverage efficiency.
Incident Flow Timeline	Time-series of 911 calls color-coded by incident type.	Highlights patterns and surges across different categories and time periods.

Table 2: Proposed interactive visualizations for the Seattle 911 Triage/Planner dashboard.

Week — Focus	All-team activities (all five members contribute each week)
W1 (Oct 5–11) — Data audit & proposal prep	Explore Seattle 911 dataset, clean timestamps/geospatial fields, finalize scope, and assign roles. Draft and submit the project proposal by Oct 11 .
W2 (Oct 12–18) — Feature engineering & baselines	Build preprocessing for temporal and spatial features, train initial XGBoost baseline, and document results in shared GitHub repo.
W3 (Oct 19–25) — Model refinement & EDA visuals	Tune LightGBM with temporal validation, generate hotspot maps, and review related papers collaboratively.
W4 (Oct 26–Nov 1) — Optimization & integration	Implement K-Means clustering + RL prototype for resource allocation; integrate preliminary outputs into dashboard; prepare draft progress report due Oct 31 .
W5 (Nov 2–8) — Explainability & interface polish	Merge predictive and optimization modules; add SHAP explainability and “what-if” interaction sliders; refine dashboard UX.
W6 (Nov 9–15) — Evaluation & results synthesis	Evaluate model accuracy (MAE, fairness, response-time reduction); finalize quantitative analyses; begin poster and report drafting.
W7 (Nov 16–21) — Final integration & submission	Freeze codebase, complete poster, README, and demo video; rehearse presentation; submit final deliverables.

Table 3: Seven-week project timeline (Oct 5–Nov 21) with proposal and progress report milestones.

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